Learning Mandarin Phonetic Representations in Neural Networks Using Acoustic Alignment

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1 Motivation

Phonetics examines the production, acoustic properties, and perception of speech sounds [1]. Different speech sounds may exhibit varying degrees of phonetic similarity, with some pronunciations being closer to each other than others. Experimental studies demonstrate that human listeners distinguish such differences when comparing spoken forms and individual phonemes [2, 3]. Graded phonetic similarity therefore describes an aspect of phonetic structure that extends beyond the identification of individual sounds. Modeling phonetic information requires investigating not only whether a system correctly identifies pronunciations but also whether its learned representations reflect the relationships between different pronunciations.

We consider the learning of phonetic representations from written Chinese character sequences, drawing inspiration from the relationship between orthographic and phonological processing in human reading. Phonological information is activated during Chinese character recognition, even when readers perform tasks involving semantic judgments [4, 5]. Mandarin provides a convenient setting for character-level pronunciation modeling, as each written character typically corresponds to a single spoken syllable in a given context. Syllables can be further decomposed into three phonological components: an initial, a final, and a lexical tone [6]. Different syllables may share one or more phonological components, whereas distinct characters may have identical pronunciations, resulting in substantial *homophony*. Furthermore, some characters have multiple pronunciations whose selection depends on the surrounding linguistic context. Learning pronunciation from written characters therefore requires a model to extract phonetic information from inputs that also carry orthographic, semantic, and contextual information. These linguistic properties may influence the organization of learned representations independently of phonetic similarity. The central challenge is consequently to determine whether a neural network can acquire representations organized according to phonetic relationships despite the presence of other linguistic information in its inputs.

The mapping from character-based representations to pronunciation can be learned using different forms of phonetic supervision. To encode target pronunciations, we adopt Hanyu Pinyin (hereafter Pinyin), a standardized Romanization system for Mandarin Chinese [7]. Rather than predicting complete romanized

syllables, we treat their initial, final, and tone components as separate categorical labels. Let y_{ini}, y_{fin}, and y_{tone} denote the ground-truth component labels, and let p_{ini}, p_{fin}, and p_{tone} denote their respective predicted probability distributions. The discrete prediction loss is defined as the sum of categorical cross-entropy terms in (1):

$$\begin{aligned} \mathcal{L}_{\text{pinyin}} = & -\log p_{\text{ini}}(y_{\text{ini}}) \\ & -\log p_{\text{fin}}(y_{\text{fin}}) \\ & -\log p_{\text{tone}}(y_{\text{tone}}). \end{aligned} \tag{1}$$

Supervised classification aims to predict these labels without explicitly specifying the geometry of the learned representation space. Although shared phonological components may induce relationships between representations, accurate classification does not necessarily imply the preservation of graded similarity between pronunciations. An alternative approach introduces *continuous acoustic representations* as training targets to guide the organization of the learned space. Let $\mathbf{h}, \mathbf{a} \in \mathbb{R}^d$ denote the learned hidden representation and its corresponding acoustic target, respectively. The acoustic alignment loss is defined as the mean squared error in (2):

$$\mathcal{L}_{\text{acoustic}} = \frac{1}{d}\|\mathbf{h} - \mathbf{a}\|_2^2. \tag{2}$$

One approach to obtaining continuous acoustic representations is to extract embeddings from pretrained speech models. For example, wav2vec 2.0 learns latent representations from speech waveforms [8]. The resulting acoustic representations encode speech information as continuous vectors rather than categorical identities, allowing relationships between pronunciations to be expressed through distances in the acoustic space. However, speech-derived representations may encode additional properties that are not directly related to the phonetic distinctions under investigation. Consequently, acoustic alignment provides an additional source of supervision but does not establish that the learned geometry corresponds to human-perceived phonetic similarity.

For independent evaluation, we employ DIMSIM, a Mandarin phonetic similarity framework [9]. DIMSIM represents Pinyin syllables through their initial, final, and tone components and calculates phonetic distance by combining distances between the corresponding components. Distances between initials and finals are obtained from learned multidimensional representations informed by linguistic structure and native-speaker similarity annotations, while tonal differences are incorporated through a separate distance component. The resulting distance measure captures graded relationships between Mandarin pronunciations and provides a reference independent of the acoustic targets used during training. By comparing DIMSIM distances with pairwise hidden-state similarities, we assess whether the learned representations preserve relative phonetic relationships beyond categorical identity. Although DIMSIM does not constitute an absolute measure of human-perceived phonetic similarity, the framework

provides a perceptually informed reference for evaluating phonetic relationships in the learned representation space.

Overall, we compare the representation spaces learned under two principal training regimes: discrete label supervision (3), where the model predicts Pinyin components, and guided supervision (4), where discrete prediction is supplemented with continuous speech-derived targets. To investigate the interaction between the training objectives, we vary the strength of acoustic alignment λ from the label-only regime toward an acoustic-dominated regime, with acoustic-only training considered as a separate reference condition (5).

$$\mathcal{L} = \mathcal{L}_{\text{pinyin}}. \tag{3}$$

$$\mathcal{L} = \mathcal{L}_{\text{pinyin}} + \lambda \mathcal{L}_{\text{acoustic}}. \tag{4}$$

$$\mathcal{L} = \mathcal{L}_{\text{acoustic}}. \tag{5}$$

The primary objective is to assess whether graded phonetic similarity emerges under discrete supervision and how continuous acoustic guidance influences its preservation in character-based representations.

2 Expected Results

The comparison of training regimes allows for several possible outcomes, each with different implications for the acquisition of phonetic structure. Under discrete Pinyin supervision, the presence of graded phonetic relationships would suggest that phonetic similarity can emerge without explicit similarity-based supervision, whereas their absence would indicate that accurate label prediction does not necessarily imply the preservation of phonetic similarity. Introducing acoustic alignment may strengthen the correspondence between hidden-state similarity and DIMSIM phonetic similarity, although the correspondence may also remain unchanged, decrease, or vary non-monotonically as the alignment strength increases. The acoustic-only regime further allows us to examine whether removing discrete supervision strengthens, preserves, or weakens graded phonetic relationships compared to the combined and discrete-only regimes. Together, the possible outcomes provide insight into how discrete pronunciation prediction and continuous acoustic guidance influence the organization of phonetic information within character-based representations.

References

1. W. J. Hardcastle, J. Laver, and F. E. Gibbon, Eds., *The Handbook of Phonetic Sciences*, 2nd ed. Chichester, U.K.: Wiley-Blackwell, 2010, doi: 10.1002/9781444317251.
2. U. Hahn and T. M. Bailey, "What makes words sound similar?" *Cognition*, vol. 97, no. 3, pp. 227–267, 2005, doi: 10.1016/j.cognition.2004.09.006.

3. T. M. Bailey and U. Hahn, "Phoneme similarity and confusability," *J. Memory Lang.*, vol. 52, no. 3, pp. 339–362, 2005, doi: 10.1016/j.jml.2004.12.003.
4. C. A. Perfetti and S. Zhang, "Very early phonological activation in Chinese reading," *J. Exp. Psychol. Learn. Mem. Cogn.*, vol. 21, no. 1, pp. 24–33, 1995, doi: 10.1037/0278-7393.21.1.24.
5. X. Zhou and W. Marslen-Wilson, "Phonology, orthography, and semantic activation in reading Chinese," *J. Memory Lang.*, vol. 41, no. 4, pp. 579–606, 1999, doi: 10.1006/jmla.1999.2663.
6. K. D. Neergaard and C.-R. Huang, "Mandarin Chinese syllable structure and phonological similarity: Perception and production studies," in *The Cambridge Handbook of Chinese Linguistics*, C.-R. Huang, Y.-H. Lin, I.-H. Chen, and Y.-Y. Hsu, Eds. Cambridge, U.K.: Cambridge Univ. Press, 2022, pp. 245–274, doi: 10.1017/9781108329019.015.
7. Y.-H. Lin, *The Sounds of Chinese*. Cambridge, U.K.: Cambridge Univ. Press, 2007, ch. 6, pp. 123–135.
8. A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 12449–12460.
9. M. Li, M. Danilevsky, S. Noeman, and Y. Li, "DIMSIM: An accurate Chinese phonetic similarity algorithm based on learned high dimensional encoding," in *Proc. 22nd Conf. Computational Natural Language Learning (CoNLL)*, Brussels, Belgium, Oct. 2018, pp. 444–453, doi: 10.18653/v1/K18-1043.